

Course Code: CS211	Course Name: Discrete Structures
Instructor Name : Jalaluddin Qureshi	
Student Roll No:	Group:

Instructions:

- Return the question paper. Read each question completely before answering it. There are **5 questions and 1 page**.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- Invigilators/ instructor can not assist you in understanding the question.
- All the answers must be solved according to the sequence given in the question paper.
- Marks will be awarded iff justifications has been provided.

Time: 60 minutes.

Max Marks: 10mark/question x 5questions = 50 marks

Part A (Set Theory)

Question 1:

Using the following relationship: $|A \cup B| = |A| + |B| - |A \cap B|$ (1),

Show that the following can be derived:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad (2).$$

Based on this exercise, and using Equation (1) and/or (2) derive the formula for $|A \cup B \cup C \cup D|$.

Solution:

Use substitution method, let, $T = B \cup C$, so we have,

$$|A \cup T| = |A| + |T| - |A \cap T|$$

$$|A \cup B \cup C| = |A| + |B \cup C| - |A \cap (B \cup C)|$$

$$\text{Since } |B \cup C| = |B| + |C| - |B \cap C|,$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |B \cap C| - |A \cap (B \cup C)|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |B \cap C| - |(A \cap B) \cup (A \cap C)|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |B \cap C| - (|A \cap B| + |A \cap C| - |A \cap B \cap C|)$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

The same approach of substitution can be used to show that,

$$|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| - |B \cap C| - |A \cap D| - |B \cap D| - |C \cap D| + |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| - |A \cap B \cap C \cap D|$$

Question 2:

If $A = \{2, 3, 4, g, \text{FAST}, \text{star}\}$, $B = \{4, f, \text{yoyo}, 2, \text{golf}, 2\}$, $C = \{\text{yoyo}, 10, 7, f, 4, \text{FAST}\}$,

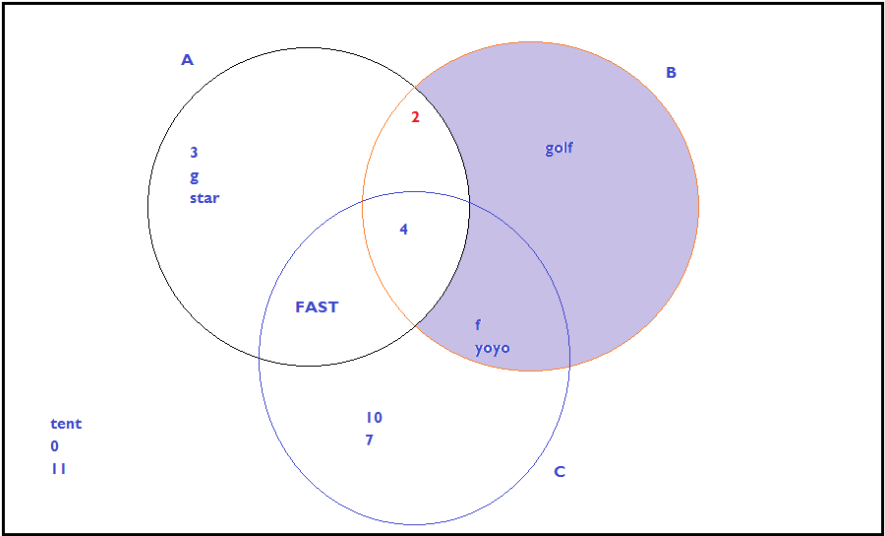
$U = A \cup B \cup C \cup \{3, \text{golf}, 7, \text{tent}, 0, 11\}$, (Hint: U is the universal set). Find the following, and plot these on a Venn Diagram (by shading appropriate region).

- (a) $A^c \cap B$ (b) $(U \setminus B) \cap C^c$
 (c) $(A \setminus U) \cup (C \setminus B)$ (d) $(A \cap C)^c \setminus B$

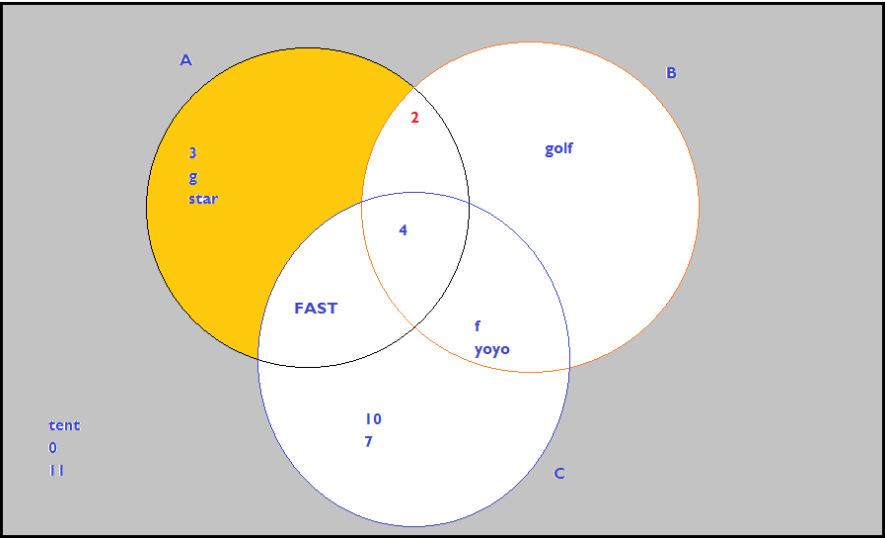
Solution:

The solution is given by the elements in the common region.

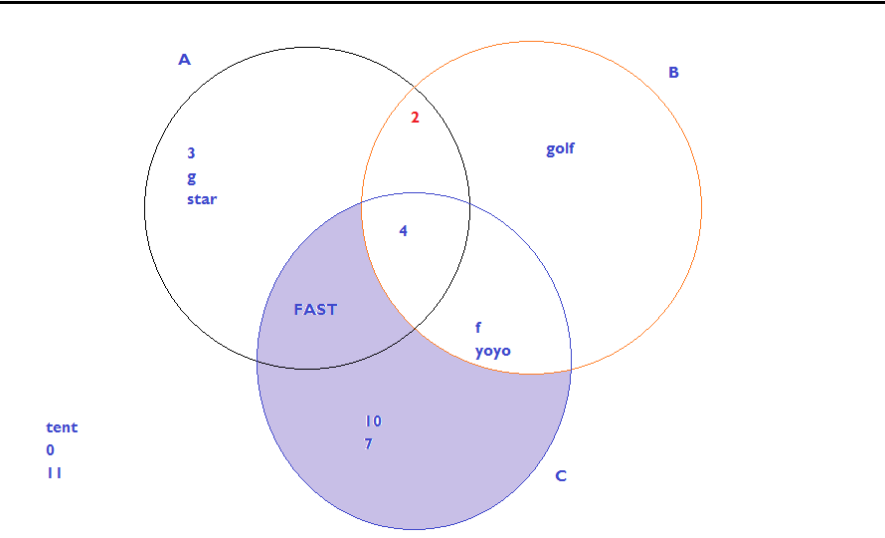
(a)



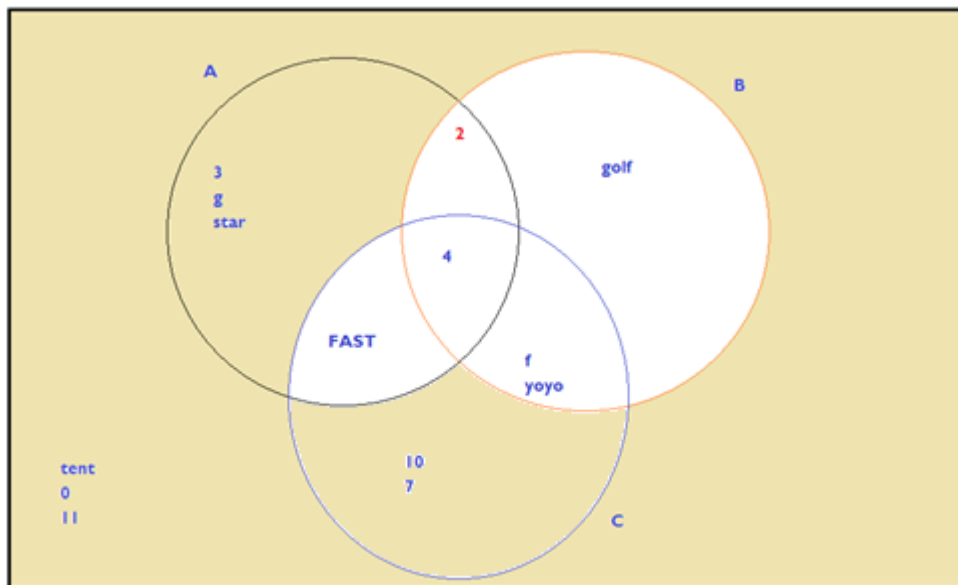
(b)



(c)



(d)



Part B (Logic Theory)

Question 3:

Determine (using appropriate technique) whether the following relationship is correct/ incorrect:

$$P \rightarrow (Q \vee R) \equiv (P \rightarrow Q) \wedge (\neg P \rightarrow R).$$

Solution:

	P	$\neg P$	Q	R	$Q \vee R$	$P \rightarrow (Q \vee R)$	M	N	$M \wedge N$
1	T	F	T	T	T	T	T	T	T
2	T	F	T	F	T	T	T	T	T
3	T	F	F	F	F	F	F	T	F
4	T	F	F	T	T	T	F	T	F
5	F	T	T	F	T	T	T	F	F
6	F	T	F	T	T	T	T	T	T
7	F	T	F	F	F	T	T	F	F
8	F	T	T	T	T	T	T	T	T

As the two truth tables are not same, the equivalence relationship is false.

As there are two tables (with total 16 entries), at the discretion of the examiner, you will lose 1 marks for a wrong entry.

Question 4:

Determine whether the assignment $c \leftarrow c+1$ (read as, c is assigned the value given by c plus one) will be executed by the if-statement, where $x < 5$, $y < 3$, $z < 7$.

- (a) If $\neg\{(x < y) \wedge (y \leq z)\}$ then $c \leftarrow c+1$ (b) If $\neg\{(x = z) \wedge ((x \leq z))\}$ then $c \leftarrow c+1$
 (c) If $\neg\{(x \geq y) \vee (x < z)\}$ then $c \leftarrow c+1$ (d) If $\neg\{(x \leq z) \vee (y = z)\}$ then $c \leftarrow c+1$

Solution:

- (a) $\neg\{(x < y) \wedge (y \leq z)\}$ then $c \leftarrow c+1$ TRUE
 $(x < y)$ is false, $(y \leq z)$ is true, their conjunction is false, and negation of this is true.
 (b) $\neg\{(x = z) \wedge ((x \leq z))\}$ then $c \leftarrow c+1$ TRUE
 $(x = z)$ is false, $(x \leq z)$ is true, their conjunction is false, and negation of this is true.

- (c) $\neg\{(x \geq y) \vee (x < z)\}$ then $c \leftarrow c+1$ FALSE
 $(x \geq y)$ is true, $(x < z)$ is also true, their disjunction is true, and its negation is false.
- (d) $\neg\{(x \leq z) \vee (y = z)\}$ then $c \leftarrow c+1$ FALSE
 $(x \leq z)$ is true, $(y = z)$ is false, their disjunction is true, and its negation is false.

Question 5:

Using the algebra of Propositions, simplify the following:

$$(x \vee \neg y \vee z) \wedge (\neg x \vee \neg y \vee z) \vee (w \wedge \neg x)$$

After the simplication, draw its truth table.

Solution

$$(x \vee \neg y \vee z) \wedge (\neg x \vee \neg y \vee z) \vee (w \wedge \neg x)$$

Conjunction has the highest precedence so,

$$= \{ [x \wedge (\neg x \vee \neg y \vee z)] \vee [\neg y \wedge (\neg x \vee \neg y \vee z)] \vee [z \wedge (\neg x \vee \neg y \vee z)] \} \vee (w \wedge \neg x)$$

$$= \{ \cancel{[x \wedge \neg x]} \vee \cancel{[x \wedge \neg y]} \vee [x \wedge z] \} \vee \{ \cancel{[\neg y \wedge \neg x]} \vee \cancel{[\neg y \wedge \neg y]} \vee [\neg y \wedge z] \} \vee \{ \cancel{[z \wedge \neg x]} \vee \cancel{[z \wedge \neg y]} \vee [z \wedge z] \} \vee (w \wedge \neg x)$$

$$= \cancel{[x \wedge \neg y]} \vee [x \wedge z] \vee \cancel{[\neg y \wedge \neg x]} \vee \cancel{[\neg y \wedge \neg y]} \vee [\neg y \wedge z] \vee \cancel{[z \wedge \neg x]} \vee \cancel{[z \wedge \neg y]} \vee [z \wedge z] \vee (w \wedge \neg x)$$

$$= \cancel{[x \wedge z]} \vee \cancel{[\neg y \wedge z]} \vee [z \wedge z] \vee (w \wedge \neg x) = \bar{y} \vee z \vee (w \wedge \neg x)$$

It is possible to simplify it further, but the above solution is sufficient, given time limitation in exam.

Based on the above reduction, a truth table for the above can be easily drawn.